

CISC 227 A1

Comparison Write-Up

Human vs. AI Requirements Analysis

Comparing the human-generated stakeholder requirements with the AI-generated requirements revealed distinct differences in focus, specifically regarding business logic versus backend technical logic. The human role-play process was essential for surfacing real-world business constraints and stakeholder conflicts that the AI failed to anticipate. Conversely, the AI provided a highly structured baseline for database management and system state transitions that the human actors overlooked.

What the AI Missed

The AI generated a generic inventory system and completely missed the nuanced business operations of a rental shop. Most notably, it failed to identify the core stakeholder conflict regarding monetary deposits, where the Renter opposed deposits to reduce friction, but the Clerk demanded them to secure high-value inventory. The AI also missed usability and operational constraints, such as the Clerk's need for a rapid checkout process (under 3 seconds and minimal screens) and the Owner's requirement for a graphical user interface to avoid command-line administration.

What the AI Added

While the human requirements focused heavily on user experience and business rules, the AI excelled at defining the foundational backend logic necessary for the system to function securely. The AI successfully introduced critical technical requirements, such as generating unique rental transaction IDs, timestamping all events, and implementing state-locking mechanisms to prevent items marked as "rented" from being checked out concurrently.

Conclusion

Neither list was sufficient on its own. The final refined requirements list required synthesizing both sources: treating the AI's list as the technical framework and using the human-generated list to enforce business rules, dictate user interfaces, and resolve operational conflicts.

Final Refined Requirements List

- The system shall allow users to view the current availability status of specific gear items via a public-facing web interface.
- The system shall process gear checkouts using a customer's name and phone number as the unique identifier.
- The system shall process rentals without requiring the creation of a password-protected user account.
- The system shall mandate a valid ID and a monetary deposit hold for equipment with a replacement value over \$100.
- The system shall generate a printed or emailed receipt listing borrowed items and their due dates at checkout.
- The system shall send an SMS reminder to the customer 24 hours before the rental period expires.
- The system shall allow staff to extend an active rental period by 1-2 days via front-desk intervention.
- The system shall return search results for gear availability and current borrower information in under 3 seconds.
- The system shall allow staff to search for rental records using a customer's name.
- The system shall allow staff to search for inventory using the gear's name rather than its exact ID number.
- The system shall allow staff to complete a checkout process using no more than 3 data entry screens.
- The system shall allow staff to flag returned inventory as "out of service" if it is damaged.
- The system shall prevent items flagged as "out of service" from being checked out to a new customer.
- The system shall track and report the most frequently rented items by month.
- The system shall generate reports identifying currently checked-out gear and overdue items.

- The system shall allow staff to manually toggle a 'Trusted Regular' tag on a customer's profile.
- The system shall allow staff to append text-based condition reports to a gear item's profile.
- The system shall generate a unique rental transaction ID for each checkout event.
- The system shall timestamp all rental and return events.
- The system shall automatically flag a rental as overdue when the current date exceeds the due-back date.